

# MUHAMMAD ARHAM

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## Skills Summary

**Languages:** Python, C++

**ML Frameworks:** PyTorch, JAX, Tensorflow, HuggingFace, LangGraph, Accelerate, DeepSpeed, TRL, Unsloth, TensorRT

**Systems & Backend:** FastAPI, Git, SQL & NoSQL, Vector Databases, Redis, Celery, Docker, AWS, CI/CD

**Research Interests:** Deep Reinforcement Learning, LLM Post-Training, Continual Learning, Test-Time Training

## Education

**University of Alberta**

Sep 2025 - Present (*Expected 09/27*)

*MS Computing Science, Supervisor: Dr. Martha White ☞, RLAIM, Amii*

**National University of Sciences & Technology (NUST)**

Sep 2020 - May 2024

*BE Software Engineering, CGPA 3.86 / 4.00, MITACS & CERN Intern, Rector's Gold Medal*

## Selected Publications

**Directed Exploration in Actor-Critic Methods via Ensemble-Based Value Bonuses**

In Progress

*Muhammad Arham, Aditya Soni, Sam Scholnick-Hughes, Martha White*

**LLM-Informed Discrete Prompt Optimization ☞**

LLM Cognition Workshop @ ICML

*Muhammad Arham\*, Zeeshan Memon\*, Adnan-ul-Hasan, Faisal Shafait*

*Published*

**Alif: Advancing Urdu LLMs via Multilingual Synthetic Data Distillation ☞**

MRL @ EMNLP

*Ali Shafique, Kanwal Mehreen, Muhammad Arham*

*Published*

## Industrial Experience

**Google DeepMind**

May 2026 - Present

*Research Software Developer*

*Montreal, Canada*

- Collaborating with the FAI Incubation team to support external partners on early-stage frontier AI projects.

**ServeDen**

Apr 2025 - Dec 2025

*Machine Learning Engineer*

*Remote, USA*

- Built a no-code LLM agent provisioning platform for deploying, managing, and executing agentic applications.
- Developed a modular agentic LLM framework with support for multi-agents, MCP integrations, guardrails, & streaming.
- Implemented end-to-end RAG infrastructure, covering document ingestion, hybrid retrieval, re-ranking, and evaluation.
- Designed a graph-based Agent Builder enabling workflow creation via nodes and edges with stateful workflow execution.

**Traversaal.ai ☞**

Sep 2024 - Feb 2025

*NLP Researcher*

*Remote, USA*

- Implemented a production-grade platform combining *RAG pipelines* with *agentic multi-modal systems* for context-aware information retrieval, report synthesis, and automated generation of insights from proprietary data.
- Provided B2B consultancy to clients on RAG systems (>\$200K contracts), including Ikona Analytics, JIB, & Kraft.
- Core member of development of *Urdu-Llama* under Traversaal.ai's LARGE initiative, designing scalable training and evaluation pipelines for low-resource language modeling; won the META AI APAC Accelerator Grant (USD 100K).

**CERN ☞**

June 2024 - Sep 2024

*CERN Summer Student*

*Meyrin, Switzerland*

- Contributed to the CMSSW project under the Patatrack team at CERN, for the next phase of HL-LHC.
- Profiling and optimizing heterogeneous CUDA code for the event data reconstruction of the TICL project for CMS.
- Porting CPU-only algorithms to heterogeneous CUDA and ROCM devices using the alpaka framework.
- Official report titled *Optimization of Event Reconstruction within the TICL Framework in CMSSW* available on CDS ☞

**Vyro ☞**

Sep 2022 - Apr 2024

*Machine Learning Engineer*

*Islamabad, Pakistan*

- Developed and maintained PhotoTune, ImagineAI, Chatly, ReShot, ReTouch and RemoveIt for millions of users.
- Optimized Stable Diffusion Generative AI pipelines, resulting in a *fivefold acceleration in inference speed and a 50% reduction in GPU consumption, using torch compilation, ONNX, TensorRT, and Oneflow computational graph caching*.
- Implemented and deployed custom image generation pipelines using Diffusers and ComfyUI for inpainting, outpainting, face-swapping, and avatar creation within the ImagineAI application. Integrated support for SD1.5, SD2.1, and SDXL.
- Trained and deployed image segmentation models, including DeepLabV3 and U2Net, for image editing application.
- Enhanced GAN-based upscaling models, including SwinIR & ESRGAN, for improved image super-resolution.
- Implemented a multi-modal RAG pipeline for document-based question answering with HuggingFace and PineconeDB.